NOT MEASUREMENT SENSITIVE

MIL-STD-361-4(TM)
15 JULY 1992

SUPERSEDING
(See 6.5.)

MILITARY STANDARD

MANUALS, TECHNICAL:

QUALITY ASSURANCE REQUIREMENTS

AMSC A6753

AREA TMSS

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FOREWORD

1. This military standard is approved for use by the Department of the Army and is available for use by all Departments and Agencies of the Department of Defense (DOD).

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, USAMC Materiel Readiness Support Activity, ATTN: AMXMD-EP, Lexington, Kentucky 40511-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

3. This document is one part of an 11-part bookform standard to be used in the preparation and acquisition of technical manuals that are essential to support (operate and maintain) the various types of equipment and weapon systems within the Department of the Army.

4. This document contains technical manual quality assurance requirements for the preparation of technical manuals.

5. Other parts of this multipart standard are:

MIL-STD-361-1	-	Writing Style and Format Requirements.
MIL-STD-361-2	-	Comprehensibility Requirements.
MIL-STD-361-3	-	Requirements for Graphics.
MIL-STD-361-5	-	Requirements for Packaging.
MIL-STD-361-6	-	Introductory Information with Theory of Operations Requirements.
MIL-STD-361-7	-	Operator and Preventive Maintenance Instructions.
MIL-STD-361-8	-	Maintenance Instructions.
MIL-STD-361-9	-	Troubleshooting Procedures.
MIL-STD-361-10	-	Lubrication Instructions.
MIL-STD-361-11	-	Repair Parts and Special Tools List (RPSTL).

6. By incorporating standardized and service-unique technical manual requirements, this document and its dash-numbered companion parts, along with the general acquisition specification MIL-M-63008, Acquisition/Preparation Requirements, support the acquisition of technical manuals.

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1. SCOPE

1.1 Scope. This document establishes technical manual quality assurance (TMQA) requirements necessary to ensure the adequacy and accuracy of technical manual (TM) content through a coordinated contractor and Government program. Procedures are established for the development and implementation of a contractor program encompassing the conduct of TMQA program reviews, quality reviews, in-process reviews, validation, and verification requirements.

1.2 Purpose. This TMQA document supplements the various format and content requirements in the other parts to this 11-part standard for specific types of technical manuals and related publications and does not alone deliver any technical data.

1.3 Applicability. This document applies to all TM acquisitions when referenced in the contract or other contractual document. This document is applicable for use by all the Department of the Army.

1.4 Selective application and tailoring. This document contains some requirements that may not be applicable to the preparation of all technical manuals. Selective application and tailoring of requirements contained in this document are the responsibility of the contracting activity and shall be accomplished through the use of MIL-M-63008. The applicability of some requirements is also designated by one of the following statements:

- unless requirement is specifically excluded by the contracting activity; or
- as/when specified by the contracting activity.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation. (See 6.2.)

SPECIFICATIONS

MILITARY

MIL-I-45208	-	Manuals, Technical: Inspection System Requirements.
MIL-M-63008	-	Manuals, Technical: Acquisition/ Preparation Requirements.
MIL-Q-9858	-	Manuals, Technical: Quality Program Requirements.

STANDARDS

MILITARY

MIL-STD-109	-	Manuals, Technical: Quality Assurance Terms and Definitions.
MIL-STD-361-2	-	Manuals, Technical: Comprehensibility Requirements.
MIL-STD-361-11	-	Manuals, Technical: Repair Parts and Special Tools List (RPSTL).

(Unless otherwise indicated, copies of federal and military specifications and standards are available from Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5904.)

2.1.2 Other Government documents and publications. The following other Government document/publication forms a part of this document to the extent specified herein. Unless otherwise specified, the issue is that cited in the solicitation.

DOD 5010.12-L	-	Acquisition Management Systems and Data Requirements Control List.

(Application for copies should be addressed to Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5904.)

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2.2 Order of precedence. In the event of a conflict between the text of this document ("This document" includes all volumes of MIL-STD-361 series.) and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. DEFINITIONS

3.1 Terms and definitions. The quality assurance (QA) terms and definitions used in this document are consistent with those contained in MIL-STD-109 except as defined in 3.1.1 through 3.1.8.

3.1.1 Acronyms used in this standard. The acronyms used in this standard are defined as follows:

a.	AAL	-	Additional Authorization List.
b.	AMSDL	-	Acquisition Management Systems and Data Requirements Control List.
c.	BII	-	Basic Issue Items.
d.	CAGEC	-	Commercial and Government Entity Code.
e.	CD	-	Classification of Defects.
f.	DID	-	Data Item Description.
g.	DOD	-	Department of Defense.
h.	DODISS	-	Department of Defense Index of Specifications and Standards.
i.	IPR	-	In-Process Review.
j.	LSA	-	Logistic Support Analysis.
k.	MAC	-	Maintenance Allocation Chart.
l.	NSN	-	National Stock Number.
m.	PMCS	-	Preventive Maintenance Checks and Services.
n.	QA	-	Quality Assurance.
o.	RGL	-	Reading Grade Level.
p.	RPSTL	-	Repair Parts and Special Tools List.
q.	SMR	-	Source, Maintenance, Recoverability.
r.	TM	-	Technical Manual.
s.	TMQA	-	Technical Manual Quality Assurance.

3.1.2 In-process review (IPR). A review of contractual requirements, technical documentation, and TM increments which may be carried out at any time during the TM development to:

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a. Evaluate the product.

b. Ensure that the TM, supplement, change, or revision is being written according to the applicable specifications.

c. Ensure the document(s) correctly reflect the approved configurations of the appropriate hardware and software.

3.1.3 Intermediate product. Work in progress and supporting source data.

3.1.4 Quality program review. A Government evaluation of quality-related data generated by the contractor as part of the QA program. The Government evaluation determines contractor compliance with the approved TMQA program plan. Quality program reviews evaluate the contractor QA program and should not be confused with technical review of TMs (eg, IPRs).

3.1.5 Quality review. A selective comparison of development processes and products with a given set of standards or objectives.

3.1.6 Technical manual quality assurance program. A systematic, coordinated effort to establish a high level of confidence that the TM product offered conforms to established, contractually defined technical requirements. A QA program includes efforts by the contracting activity and preparing activity including, but not limited to, IPRs, validation, and verification.

3.1.7 Validation. The process by which the contractor tests a TM for technical accuracy and adequacy, comprehensibility, and usability. It is conducted at the contractor's facility, operational site, or other site, as applicable, and involves the hands-on 100 percent performance of operating and maintenance procedures including checkout, calibration, alignment, preventive maintenance checks and services, scheduled removal and installation instructions, and troubleshooting (fault isolation).

3.1.7.1 Validation/verification, combined. The final QA iteration by the Government for acceptance of the TM during which a TM is tested to determine its adequacy and operational suitability for operation and maintenance of equipment or systems. This process also includes contractor participation in testing a TM for technical accuracy and adequacy, comprehensibility, and usability. It involves the hands-on 100 percent performance of operating and maintenance procedures including checkout, calibration, alignment, preventive maintenance checks and services, scheduled removal and installation instructions, and troubleshooting (fault isolation).

3.1.8 Verification. The final QA iteration by the Government for acceptance of the TM during which a TM is tested to determine its adequacy and operational suitability for operation and maintenance of equipment or systems. Verification may be tailored based on the Government's confidence level in the preparing activity's QA program, compliance with provisions of

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the specifications and other technical contract requirements, and effective integration of logistic support requirements for the tasks to be performed. Verification is conducted using production equipment and target audience soldiers provided by the proponent schools (operator and maintenance). The contracting activity shall coordinate with the equipment's proponent school for the target audience. If there are dual proponents (operator and maintenance levels in the same TM), the contracting activity shall coordinate both.

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4. GENERAL REQUIREMENTS

4.1 Responsibility for inspection. Unless designated otherwise herein or in the contract or purchase order, the contractor shall be responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Validation is the responsibility of the contractor and verification the responsibility of the contracting activity. Unless requirement is specifically excluded by the contracting activity, the contractor shall use his own or any other facilities suitable for the performance of the inspection requirements specified herein. The Government reserves the right to perform any of the inspections set forth in this standard where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

NOTE

Reference to technical manuals (TMs) includes technical manuals, changes, revisions, and supplements.

4.1.1 Quality assurance (QA) provisions. Quality assurance in the preparation of TMs shall be the responsibility of the contractor and shall be implemented in accordance with this standard. TMs furnished in accordance with this standard shall be inspected by the supplier for conformance to the applicable requirements herein.

4.1.2 Contracting activity inspection. TMs submitted by the contractor shall be subject to inspection, review, verification, and approval or disapproval by the contracting activity as determined by the terms of the contract. Inspection and verification will be performed by the Government prior to acceptance.

4.1.3 Responsibility for compliance. All items shall meet all requirements of this standard and the statement of work. The inspections set forth in this standard shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in this standard shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection in quality conformance is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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5. DETAILED REQUIREMENTS

5.1 Technical manual quality assurance (TMQA) program. The contractor shall establish a written TMQA program in accordance with the requirements of this standard which supplements and details the requirements of MIL-Q-9858 and MIL-I-45208 to ensure the development of technically accurate and complete TMs. A TMQA program developed as a result of material generated by the requirements of MIL-Q-9858 and MIL-I-45208 will be reviewed by the contracting activity for acceptance provided it satisfies the requirements of this standard. The contractor's TMQA program shall encompass the accountability for and development of quality control functions required for the management of the following TMQA program elements:

a. Source data collection (current data).

b. Intermediate product.

c. Graphics and illustrations.

d. Validation.

e. Internal coordination.

f. Records.

g. Verification support.

h. Final product.

5.1.1 Guidance and Quality Planning Conference. The Guidance and Quality Planning Conference is conducted to ensure the contractor's understanding of applicable standards and specifications, TM contract requirements, formal instructions, established policies, and program requirements. Such conferences may be requested by either the contractor or Government.

5.1.2 TMQA program plan. Unless requirement is specifically excluded by the contracting activity, the contractor shall prepare a plan which shall describe the scope and approach of the TMQA program. The contractor shall be responsible for the implementation of the TMQA program plan and for product quality. (See 6.4.)

5.1.3 TMQA program organization. The contractor's TMQA program organization shall have well defined responsibility, authority, and the organizational freedom to identify and evaluate TMQA problems and to recommend and initiate solutions.

5.1.4 TMQA program functions. All TM elements and processes shall be evaluated by contractor and Government QA personnel at various stages of development, by any or all of the following TMQA program functions:

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a. Guidance and quality planning conferences. (See 5.1.1.)

b. TMQA program reviews. (See 5.1.7.)

c. Quality reviews. (See 5.2.)

d. In-process reviews (IPRs). (See 5.9.)

e. Validation. (See 5.10.)

f. Verification. (See 5.12.)

5.1.5 TMQA program plan acceptance. The Government will furnish written notice of the acceptability of the contractor's TMQA program plan.

5.1.6 TMQA program plan implementation. Implementation of the TMQA program plan shall be evidenced by development of:

a. Review procedures and records,

b. Corrective and preventive action procedures,

c. Support for QA functions,

d. Product validation procedures, and

e. Compliance with the approved milestone dates.

5.1.7 TMQA program reviews. During the TMQA program reviews, the contractor shall demonstrate to the Government the implementation of the TMQA program. This shall include review of data generated during contractor quality reviews and quality related reports and records. TMQA program reviews chaired by a Government representative will be conducted at the contractor's facility. All quality review results will be documented by the Government. The contractor shall support quality program reviews as requested by the Government and provide access to QA records as specified in the TMQA program plan. The Government reserves the right to conduct quality program reviews throughout the term of the contract to ensure compliance with the TMQA program plan, applicable TM standards and specifications, the contract, and the production of a quality product.

5.2 Quality reviews. During TM development and production, the contractor shall perform reviews of the TM and of all the constituent elements and processes. The reviews shall be used to assess compliance with the TMQA program plan and provide for corrective action. The contractor's QA organization shall conduct quality reviews to ascertain compliance to the requirements of this standard. Quality reviews shall be conducted to evaluate the availability and adequacy of materials, processes, procedures, and intermediate products which constitute TM development. Sampling plans (See 5.8.) shall be as specified in the TMQA program plan. Desk reviews (to include Reading Grade

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Level (RGL) checks shall be a part of the quality reviews. See 5.10.2.

5.3 Quality records. The contractor shall maintain objective records of all quality reviews as specified in the contract and the TMQA program plan. The contractor's compliance with the requirements of having objective records shall be determined by the ease of retrieval of specific information from those records and their accuracy, currency, and completeness at the time of the Government representative's request.

5.4 Corrective action. The contractor shall initiate a process of corrective action for detected deficiencies. The contractor shall implement preventive action programs to counter any apparent deficiency trends. The contractor QA personnel shall approve all corrective and preventive actions taken in response to all recorded deficiencies. Objective evidence of the effectiveness of the corrective action program for each deficiency shall be maintained.

5.5 Source data control. The contractor shall ensure that the current or contractually correct source data (as of copy freeze date) is available and utilized for TM development.

5.6 Maintenance task content and organization. The TM maintenance task content and organization shall be consistent with the Functional Group order of the Maintenance Allocation Chart (MAC). The contractor shall ensure that all required levels of maintenance are sufficiently detailed and complete. The TM maintenance tasks shall be in agreement with the logistic support analysis (LSA) task analysis data; approved maintenance breakdown; and approved source, maintenance, and recoverability (SMR) codes. Evaluation shall include a comparison of the tasks identified to the current configuration of hardware.

5.7 Control of subcontractors and vendors. The contractor shall ensure the quality of TMs prepared by subcontractors and vendors. Successful implementation of the contractor's procedures designed specifically for control of subcontractors and vendors shall be the evidence of compliance with this requirement. As a minimum, the contractor shall establish procedures for the following:

a. The selection of qualified subcontractors and vendors.

b. The transmission of quality requirements for subcontractors and vendors.

c. The inspection of subcontractor and vendor records and corrective action.

d. The evaluation of subcontractor and vendor materials, processes, and products.

5.8 Sampling plans. All TM products, regardless of percentage of completion, shall be sampled and evaluated as a method of determining the acceptability of product in development.

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Sampling plans shall be as specified in the TMQA program plan. The contractor's QA organization shall, when appropriate, utilize sampling techniques for evaluation of product.

5.8.1 Sample classification of defects (CD). The contractor shall prepare a CD list patterned after the CD list below for product evaluation. The contractor and the Government may jointly classify additional defects applicable to the specific products being procured.

a. Major defects (incorrect, incomplete, missing).

(1) Maintenance procedures,
(2) Values and tolerances,
(3) Illustrations, schematics, wiring diagrams,
(4) Part numbers,
(5) References and indexes,
(6) Safety - notes, cautions, warnings,
(7) Technical content (source data and hardware comparison),
(8) Classified matter (incorrect identification or handling),
(9) Charts and tables, and
(10) An excess of any one or combination of the following: unfamiliar words, inconsistent vocabulary, long sentences, long paragraphs, noninformative headings, organization not based on immediate needs of the user, and complex or unclear illustrations.

b. Minor defects (incorrect, incomplete, missing).

(1) Typographical errors,
(2) Collated pages,
(3) SMR codes, and
(4) Style and format (writer's guide errors).

5.9 In-process reviews (IPRs). IPRs will be authorized and convened by the Government during the preparation of the TM, primarily as guidance to the contractor and to assure that the publications are prepared in conformance with the contract and standards and specification requirements. The contractor shall support IPRs and provide access to TM materials and equipment, facilities, intermediate and final products, or other support as specified by the contracting activity. IPRs may include evaluation of source data, TM plans or outlines, presentation methods, modes of preparation, standards and specification compliance, completed text and artwork, and readability. These IPRs are not part of the validation or verification and shall not be used to take the place of validation or verification.

5.9.1 IPR location. IPRs shall be held at the contractor's facility, at a designated Government facility, or at any other site as specified by the contracting activity prior to the IPR. The contractor may request IPRs at any time during the term of the contract when assistance or clarification is desired. The

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Government will conduct additional IPRs when it appears the program is not proceeding according to schedule.

5.9.2 IPR records. The Government will act as recorder and record decisions, results, and findings during the IPRs evaluation. The Government will provide a copy of all recorded IPRs to the contractor. (See 6.4.)

5.9.3 IPR corrective actions. The Government and the contractor shall resolve IPR findings that involve problem areas or findings that require further evaluation before final disposition. Discrepancy or deficiency found as the result of the IPR shall be corrected prior to certification and acceptance of the TM.

5.10 Validation. Validation is a contractor QA responsibility which shall be accomplished for all TMs. A TM shall not be considered validated until the following conditions have been fulfilled:

a. Contractor's technical and editorial desk reviews have been performed prior to "hands on" validation. The target audience RGL cited in the contract shall be validated in conformance with the requirements of MIL-STD-361-2 prior to "hands on" validation.

b. Information reflects configuration of the systems or equipment and includes all engineering changes.

c. Procedural instructions and information are readily understandable and adequate to perform all operations and maintenance functions.

d. Adequacy and accuracy of data are checked to ensure that the data supports the approved source data, including the MAC and SMR codes of the Repair Parts and Special Tools List (RPSTL).

e. Hardware of the proper configuration is available and used for the validation effort.

f. The TM conforms to applicable requirements in the governing documents.

g. A 100 percent "hands-on" performance of all operating, maintenance, and repair procedures has been successfully performed.

5.10.1 Support equipment. Government approved support equipment shall be utilized in the performance of validation. Simulation or substitution of support equipment shall be approved by the Government. It is the responsibility of the contractor to request Government-furnished equipment in order to support the validation effort.

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5.10.2 Desk reviews. Desk reviews include the following:

a. Theory and principles of operation, system or component description, SMR codes, schematic and wiring data shall be validated against engineering source data.

b. All content shall be checked for technical accuracy and adequacy.

c. Maintenance instructions and applicable preventive maintenance checks and services (PMCS) entries shall be checked against all functions assigned in the Maintenance Allocation Chart.

d. There shall be a review for accuracy of the lists containing repair parts, special tools, and test equipment required to perform functions specified in the MAC.

e. The essentiality and adequacy of illustrations cited in the text shall be checked.

f. The existence and adequacy of references cited in the text shall be validated.

5.10.3 Validation plan. Unless requirement is specifically excluded by the contracting activity, a validation plan shall be developed and shall be acceptable to the Government. (See 6.4.) It shall outline the contractor's recommended validation procedure and indicate the scope of the validation effort. It shall also include manuals for which requirements have yet to be defined, such as equipment component and support equipment manuals. The plan shall include recommendations for simultaneous validation and verification as appropriate (5.11).

5.10.4 Validation performance of operator's, maintenance, and depot repair TMs. The contractor shall perform a 100 percent validation of all operating, maintenance, and repair procedures in the TM for technical accuracy and adequacy (exception noted in subparagraph e below).

a. As procedures are performed, this validation shall include a complete review of the associated MAC functions and levels to ensure all authorized tasks are properly included for performance at the designated level of maintenance.

b. It shall include a review of all initial setup items and references and all supply items included in the TM to ensure that they are authorized for use at the designated level.

c. This review shall include a complete survey of applicable SMR coded items to ensure that all items coded for assembly, manufacture or fabrication, replacement, or repair are included in the maintenance manual at the level designated by the SMR code. Particular care must be taken

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to ensure that the "complete repair" code (fourth position of the SMR code) is properly analyzed; this code indicates whether or not an item is to be repaired and identifies the lowest maintenance level with the capability to perform complete repair, yet does not preclude some degree of repair which may be accomplished at a lower level of maintenance. These lower level repair parts must be identified and incorporated into the maintenance procedures in the proper level TM as specified by the individual item SMR code.

d. Validation includes the actual performance by contractor personnel of operating, maintenance, and repair procedures on the equipment or system for which the TM was written. All alignment, disassembly, cleaning, inspection, testing, repair, removal, replacement, assembly, installation, troubleshooting, checkout, preventive maintenance checks and services, calibration, lubrication, and similar maintenance procedures shall be validated. All other data, such as drawings, schematics, diagrams, wiring data, and descriptive data, shall be checked against the equipment to ensure accuracy.

e. As specified by the contracting activity, a simulation of performance may be used to avoid destructive testing. Malfunctions shall only be introduced into the system or equipment when specifically required for validation of diagnostics, fault isolation procedures, or system tests. Destructive malfunctions shall not be introduced into the system or equipment for any purpose.

f. The contractor shall perform all tasks on the equipment (except as noted in subparagraph e above) with no other information than that contained in the TM. This includes common tasks such as opening hatches and removing covers, etc. Successful performance of each task will validate the technical adequacy and the clarity of that task in the TM.

g. Transitioning from one task segment or task to another, and referencing and sequencing of task segments or tasks, shall be validated by the contractor in the final TM product as a whole.

h. The contractor shall correct all errors found in the TM during the validation. The contractor shall have appropriate personnel observing the performance to determine what kind of correction is required. Minor corrections can be made on the spot and the performance continued, but no guidance or interpretation shall be provided the performer except that placed on the printed page. Numerous corrections that disrupt the performance shall be considered major corrections. In the case of major corrections of this or any type, the entire task performance shall be repeated with a corrected copy of the TM at another time.

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i. Tasks in the TM may be validated at any time. There is no requirement that they be done together or in any particular sequence. The only requirement is that the task selected for validation shall be performed completely, so that the task can be inspected for technical adequacy. No task segment which stops short of achieving the task goal shall be considered as validated.

j. The contracting activity reserves the right to witness the validation.

5.10.5 Validation performance of RPSTL. The RPSTL shall be validated by the contractor in accordance with the requirements contained in MIL-STD-361-11.

5.10.6 Separation of responsibility. The Government reserves the right to require that contractor personnel performing operating and maintenance instructions on the equipment during validation be independent from the contractor's manual preparation activity.

5.10.7 Government-furnished information. The contractor is not responsible for the adequacy and accuracy of Government-furnished information provided for inclusion in the manual(s). However, the contractor shall notify the contracting activity of any Government-furnished information which is inaccurate, inadequate, or inconsistent (with the contents of the TM, other TMs in the same series, source data, or validated/verified information). The contractor shall provide written certification to the Government that all such Government-furnished information is properly and correctly reflected in the TM.

5.10.8 Revalidation. All material changed or corrected as a result of contractor and Government testing and validation or verification shall be revalidated in accordance with the requirements of this standard.

5.10.9 Validation records. The contractor shall maintain quality assurance records that contain, at the minimum, the following information: dates of validations, tasks and material validated, method of validation (simulation or actual), findings and applicable remarks, and action taken. (See 6.4.) These records shall indicate the affected manuals, weapon system, component part number, or serial number. The contracting activity reserves the right to examine these records at the contractor's facility.

5.10.10 Correction of errors and TM validation certification. Corrections and significant comments resulting from validation shall be incorporated prior to the certification and acceptance of the TM. Unless requirement is specifically excluded by the contracting activity, the contractor shall prepare a validation certificate attesting to the TM adequacy and accuracy. (See 6.4.)

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5.11 Combined validation and verification. As specified by the contracting activity, verification shall be performed separately from or concurrently with validation. The Government retains jurisdiction over a combined validation and verification in coordination with the contractor.

5.12 Verification. Unless requirement is specifically excluded by the contracting activity, the contractor shall participate in the verification by the Government. The contractor shall provide the following types of support:

a. Record and maintain records during the verification process. The contractor shall also maintain a master copy of the TM which shall be corrected during the verification process.

b. Provide assistance to the contracting activity during the verification and provide the Government with a copy of the discrepancies revealed.

c. Make necessary corrections to discrepancies revealed during the verification process.

d. Provide the contracting activity with a report of the corrective actions taken.

e. For RPSTL verification, the contractor shall provide facilities and access for the Government to verify content accuracy of the RPSTL by inspection of the contractor's records, review of random samples of line entries, or witnessing the contractor's validation.

5.12.1 Verification methods. Representatives of the contracting activity will verify the contractor's validation to include verification of logical sequencing. The contracting activity reserves the right to perform verification by one of the methods listed in a through d below. The methods listed are in addition to reviewing the publication for conformance to the requirements of the governing documents and inspecting the contractor's validation records.

a. Performing 100 percent of the operating and maintenance procedures in the publication by using military operational and support personnel of the type and qualifications of those expected to use and maintain the equipment when deployed. Each procedure must be performed successfully at least once. The user representative must concur with the usability of the TM. All performance will be monitored by a Government subject-matter expert. A master copy of the publication being verified will be maintained to record all discrepancies. All verified material will be updated and given a final desk review. The desk review will include a check of those portions of the publication not subject to hands-on performance (e.g., index, contents, theory, basic issue items (BII),

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additional authorized items list (AAL)). The schedule followed will be that approved in the verification plan.

b. Reviewing the technical content of the publication by having Government personnel perform the operating and maintenance instructions specified by the contracting activity on the equipment, either at the contractor's plant or at Government facilities (sampling method).

c. Performing the contractor's validation at the contractor's plant, as scheduled by the contractor in accordance with the validation schedule.

d. Witnessing the validation reviews performed by the contractor.

5.12.2 Verification plan. Unless requirement is specifically excluded by the contracting activity, a verification plan shall be developed by the contractor to be reviewed and accepted by the Government. (See 6.4.)

5.12.3 Verification planning data cards. Unless requirement is specifically excluded by the contracting activity, the contractor shall develop verification planning data cards tailored to each effort. (See 6.4.) Since the verification effort is proportional to the scope of the manual and complexity of the hardware or equipment, the number of verification planning data cards may vary as follows:

a. A manual covering a single level of maintenance on a single, noncomplex item will usually require only one planning card.

b. A manual covering more than one level of maintenance will usually require planning data cards for each level of maintenance.

c. A manual covering complex equipment or systems will usually require a separate planning card for each prime heading.

5.12.4 Verification sequence control chart. Unless requirement is specifically excluded by the contracting activity, a verification sequence control chart shall be prepared by the contractor (See 6.4.) when:

a. Several manuals are to be verified under a concerted effort,

b. Several parts of a single manual are to be verified simultaneously, or

c. A preferred sequence is required to maintain continuity in task performance.

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5.12.5 Correction of errors and TM verification incorporation certification. The contractor shall incorporate all applicable comments and correct all contractor errors found in the TM during verification, at no cost to the Government. Unless requirement is specifically excluded by the contracting activity and upon completion of all verification actions, the contractor shall prepare a certificate attesting that all discrepancies and deficiencies recorded during verification have been corrected or resolved. (See 6.4.) After Government acceptance of the TM, the Government reserves the right to require the contractor to correct all errors found in the TM, at no cost to the Government, within the number of days specified in the contract. Final acceptance of the TM will be in accordance with terms of the contract.

5.13 TM evaluation records. Records shall be maintained during in-process reviews and validation and verification efforts. The records shall document quality problems and comments disposition recommendations. The records shall adequately identify the items in the manual(s) to which the comments and recommendations apply. (See 6.4.)

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This document prescribes the quality assurance requirements applicable to various types of technical manuals, revisions, supplements, and changes prepared by or for the Department of the Army.

6.2 Issue of DODISS. When this document is used in acquisition, the applicable issue of the DODISS must be cited in the solicitation. (See 2.1.1.)

6.3 Tailoring guidance. The contracting activity should tailor any required options offered herein in accordance with MIL-M-63008.

6.4 Data requirements. The following Data Item Descriptions (DIDs) must be listed, as applicable, on the Contract Data Requirements List (DD Form 1423) when this standard is applied on a contract, in order to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title
5.1.2	DI-M-2194	Technical Manual Quality Assurance Program Plan
5.10.3 5.10.9	DI-M-2195	Validation Plan
5.10.10	DI-M-2196	Validation Certification
5.9.2 5.13	DI-M-2197	Evaluation Record
5.12.2	DI-M-2198	Verification Plan
5.12.3	DI-M-2199	Verification Planning Data Cards
5.12.4	DI-M-2200	Verification Sequence Control Chart
5.12.5	DI-M-2201	Verification Incorporation Certification

The above DIDs were those cleared as of the date of this standard. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

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6.5 Supersession data. This document consolidates, standardizes, and supersedes the quality assurance requirements contained in the following documents:

MIL-M-49502(TM)	-	Manuals, Technical: Repair Parts and Special Tools List.
MIL-M-49504(TM)	-	Manuals, Technical: Combat Vehicle Precombat Checklist.
MIL-M-63004C(TM)	-	Manuals, Technical: Preparation of Lubrication Orders.
MIL-M-63036D(TM)	-	Manuals, Technical: Operators, Preparation of.
MIL-M-63038C(TM)	-	Manuals, Technical: Organizational and Aviation Unit, Direct Support or Aviation Intermediate, and General Support Maintenance.
MIL-M-63041D(TM)	-	Manuals, Technical: Preparation of Depot Maintenance Work Requirements.

6.6 Subject term (key word) listing. The following terms are to be used to identify this document during retrieval searches:

Evaluation records
Quality assurance
TMQA
TMQA program plan
Validation
Validation certificate
Validation plan
Verification
Verification incorporation certification
Verification plan
Verification Sequence Control Chart

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Army - TM

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REVIEW ACTIVITIES:

Army - AL, AR, AT, AV, CR,
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Commander
USAMC Materiel Readiness Support Activity
ATTN: AMXMD-EP
Lexington, KY 40511-5101

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INSTRUCTIONS

1. The preparing activity must complete blocks 1,2,3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4,5,6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-STD-361-4 (TM)	2. DOCUMENT DATE (YYMMDD) 920715

3. DOCUMENT TITLE
Quality Assurance Requirements

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	7. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY

a. NAME USAMC Materiel Readiness Support Activity	b. TELEPHONE *(Include Area Code)* (1) Commercial: (606) 293-3469 (2) AUTOVON: 745-3469
c. ADDRESS *(Include Zip Code)* ATTN: AMXMD-EP (TMSS) Lexington, KY 40511-5101	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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